

Exploratory Data Analysis (EDA) Report

Objective

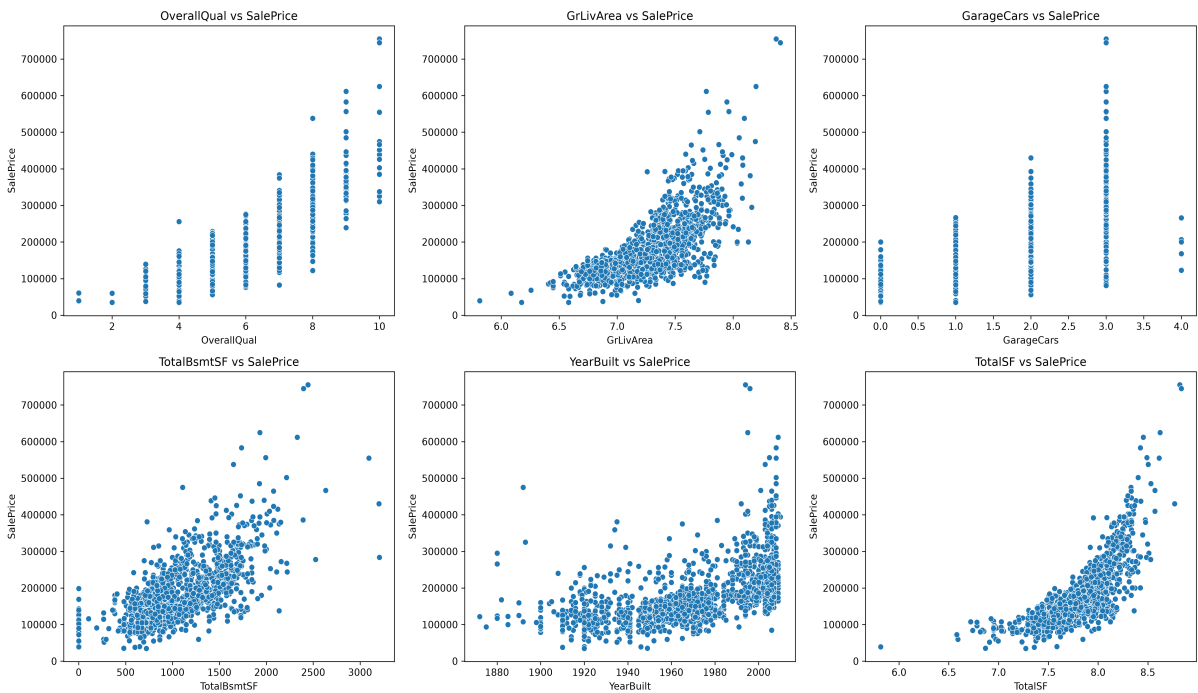
The objective of Exploratory Data Analysis (EDA) is to understand the dataset, identify patterns, detect anomalies, and analyze the relationship between the input features and SalePrice.

Dataset Overview

Dataset: House Prices – Advanced Regression Techniques
Target Variable: SalePrice
Records: 1,460
Features: 81

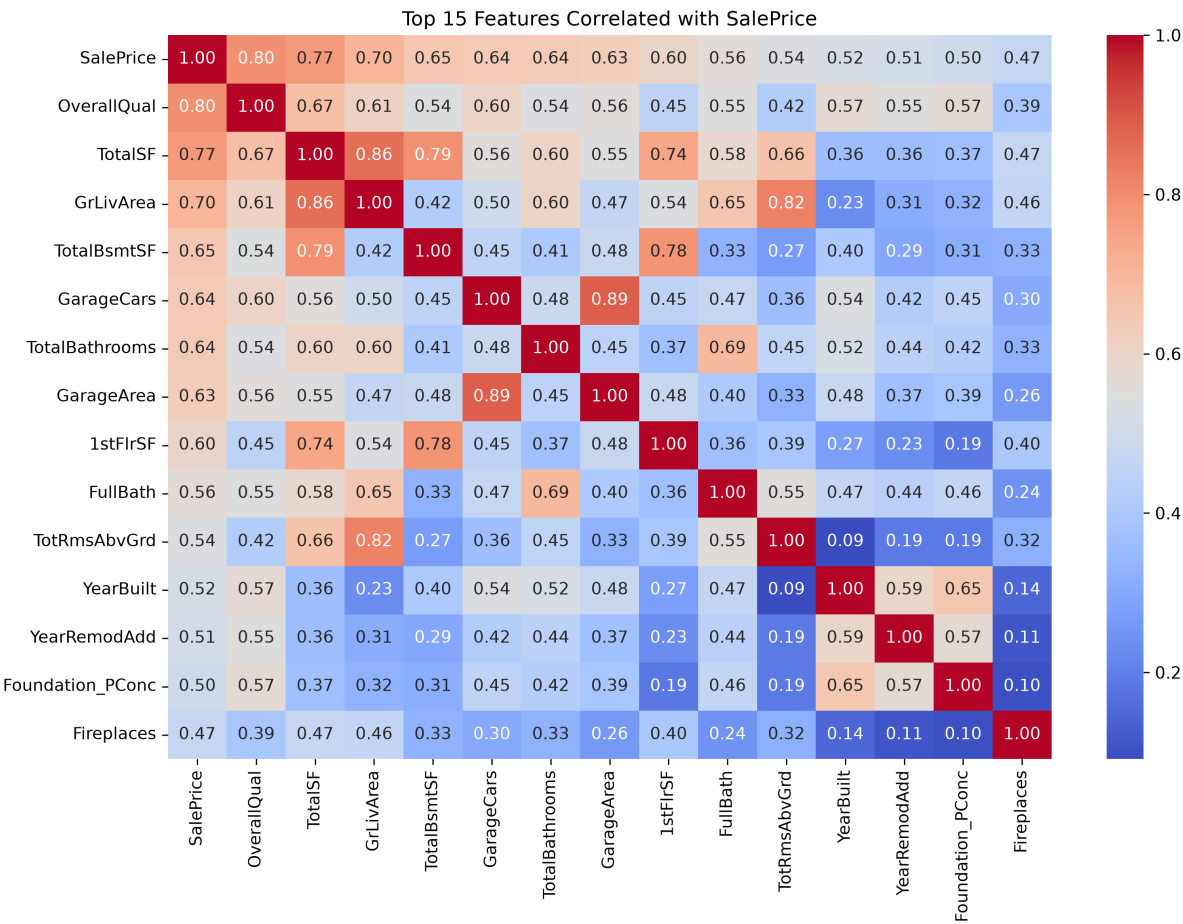
Feature vs Target Analysis

Scatter plots were used to analyze the relationship between important features and SalePrice. OverallQual, GrLivArea, GarageCars, TotalBsmtSF, YearBuilt, and TotalSF showed a positive relationship with house prices.



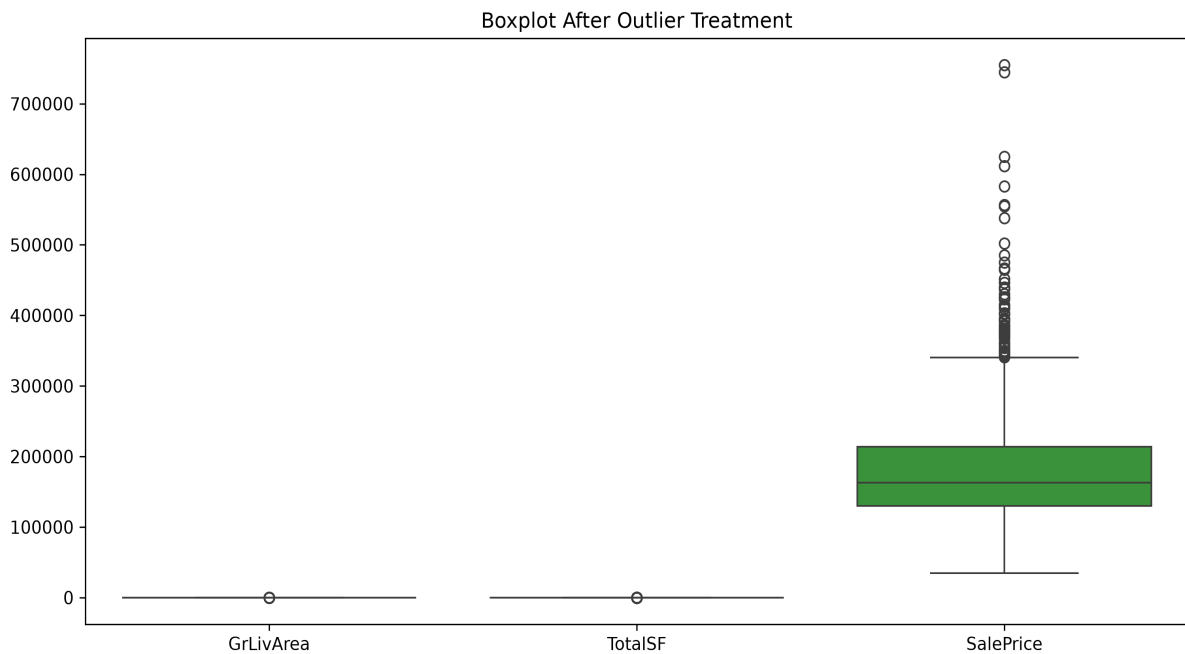
Correlation Analysis

The correlation matrix identified the features most strongly related to SalePrice. OverallQual, GrLivArea, GarageCars, and TotalSF showed the highest positive correlation.



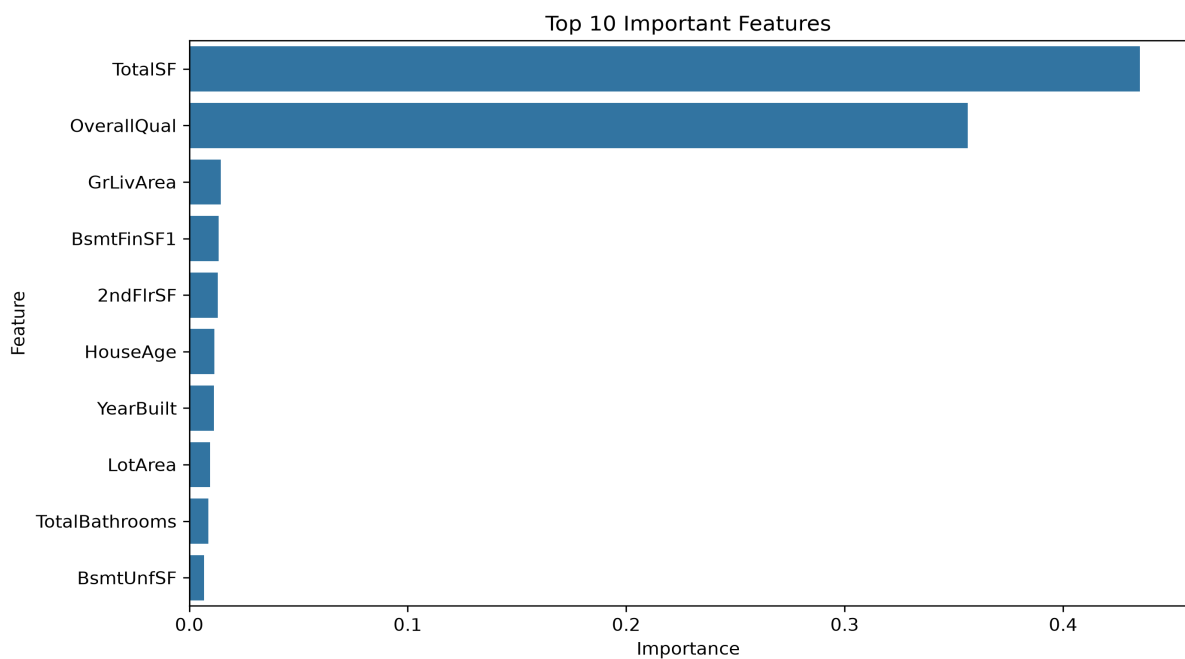
Outlier Analysis

Boxplots were used to verify that major influential outliers had been removed while preserving valid observations.



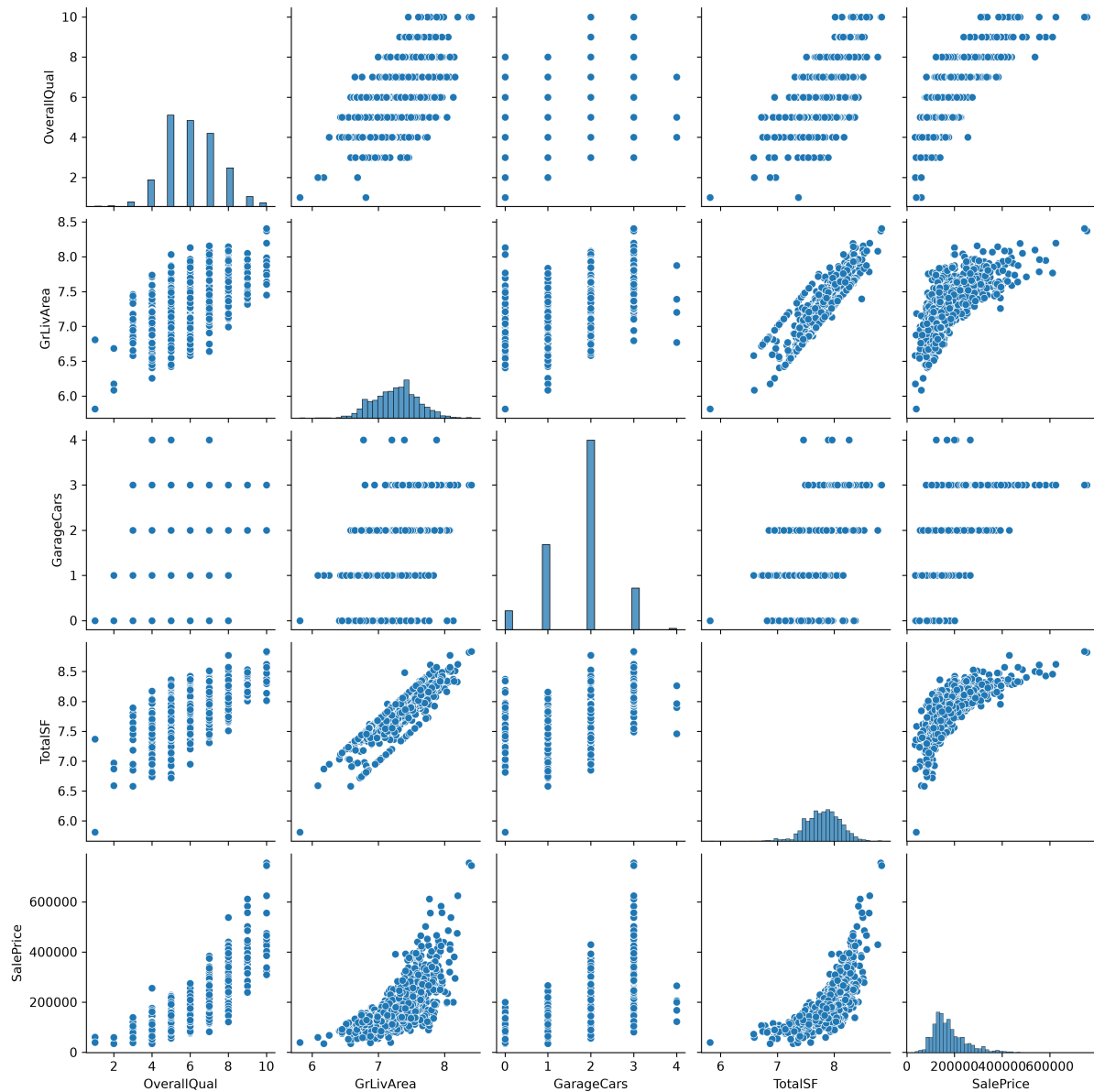
Feature Importance

Random Forest feature importance identified OverallQual, GrLivArea, TotalSF, and GarageCars as the most influential predictors.



Relationship Between Important Variables

A pair plot was created to visualize the relationships among the important features and the target variable, SalePrice.



Conclusion

EDA identified the key features influencing house prices, verified data quality after preprocessing, and provided insights that supported feature engineering and model development.